

MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY



DEPARTMENT:

ELETRONICS AND TELECOMMUNICATION
ENGINEERING

PROGRAM:

BACHELOR OF TELECOMMUNICATION
SYSTEM ENGINEERING

COURSE NAME:

FUNDAMENTAL OF ELECTRONICS

COURSE CODE:

ET 8107

TASK:

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LEVEL:

UQF 8 FIRST YEAR

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Qns 2: Design DC equivalent circuit from Transistor amplifier with two sources.

Transistor amplifier: Is the electronic device which is used to raise the strength of a weak signal in which small input voltage results in a large output circuit voltage.

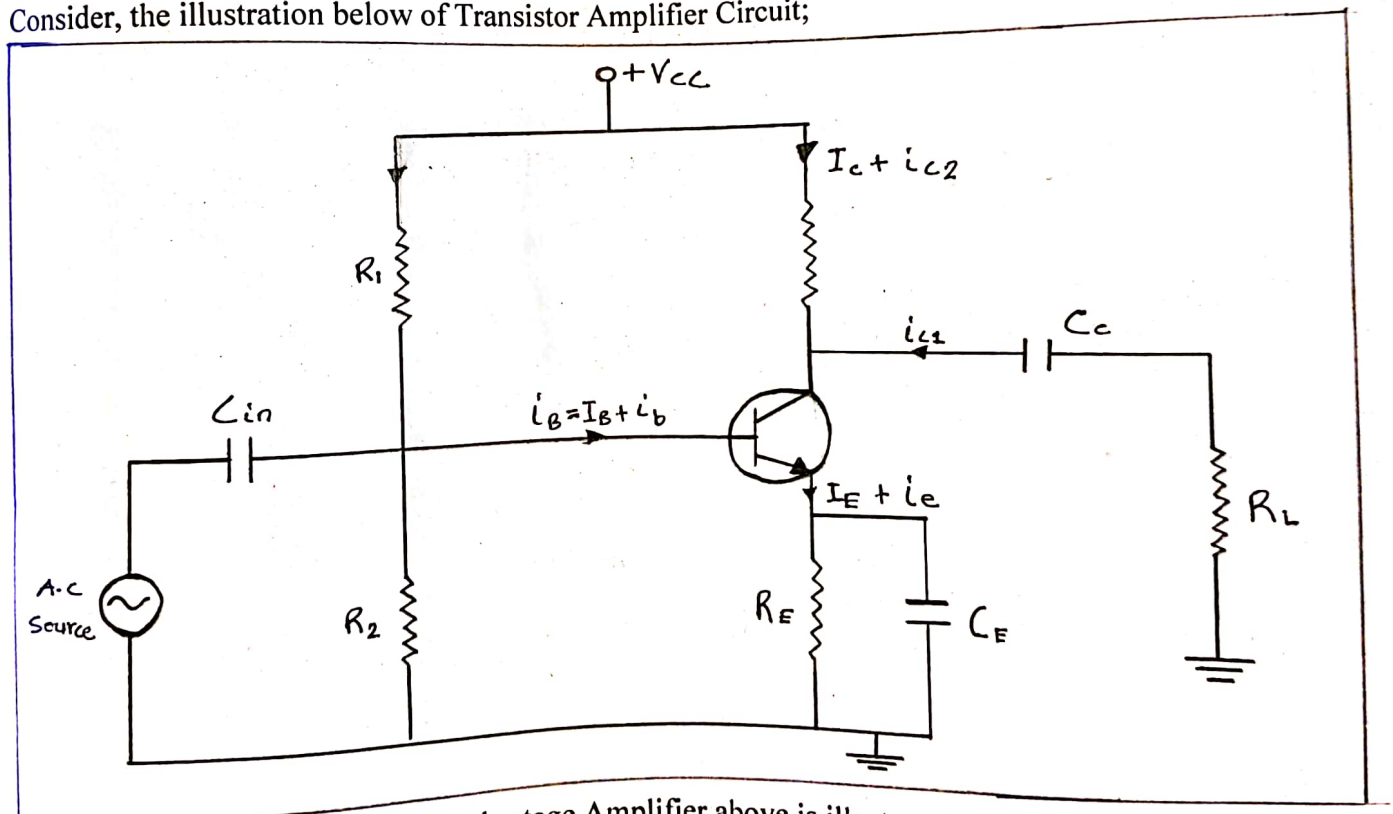
Types of Transistor Amplifier;

- i. Single stage Amplifier.
- ii. Multistage Amplifier.

Here, we are dealing with single stage Amplifier with two sources which are: -

- i. DC source (i.e. $+V_{CC}$)
- ii. AC source (i.e. $\sim$)

Consider, the illustration below of Transistor Amplifier Circuit;

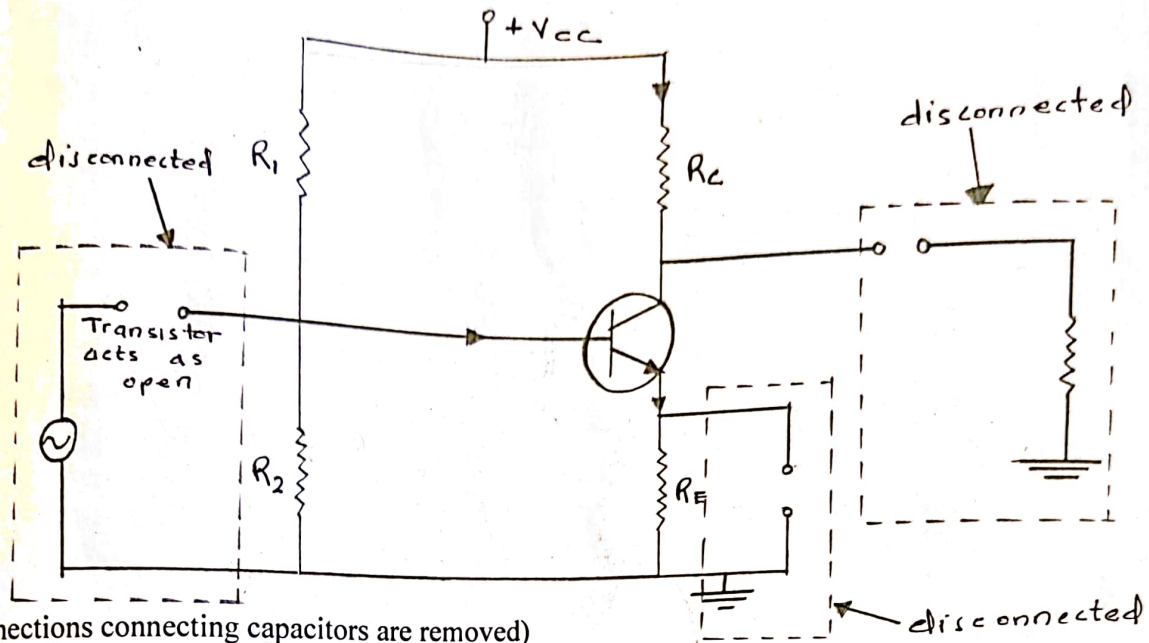


The DC equivalent circuit from single stage Amplifier above is illustrated below:

Procedure used to obtain DC equivalent circuit is;

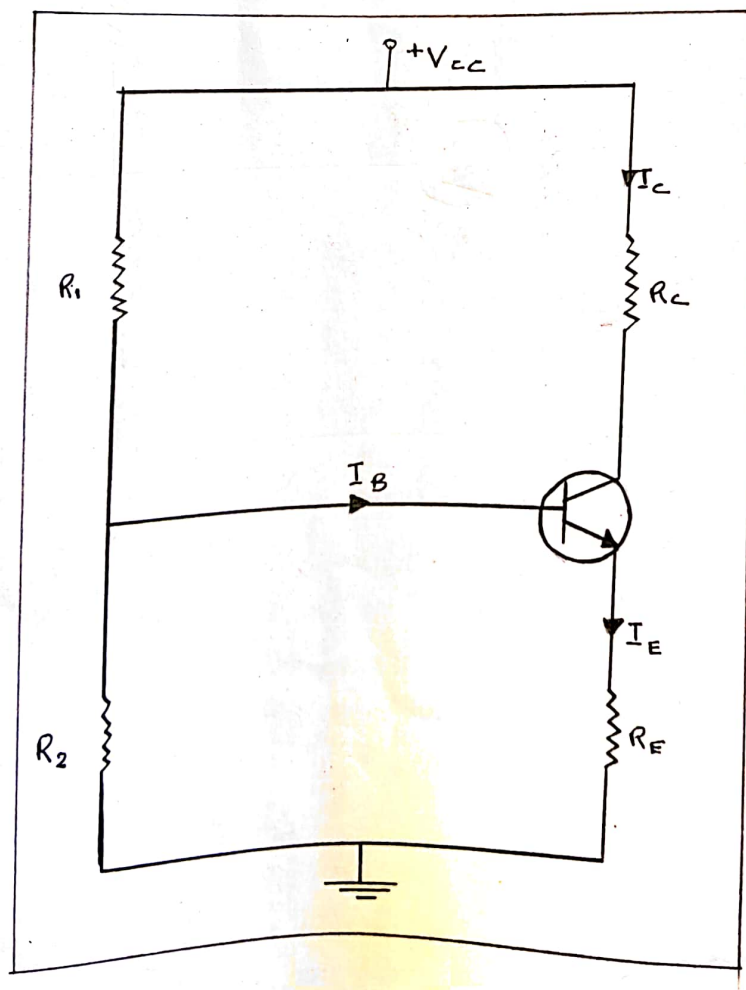
- ✓ All capacitors are replaced by open circuits.

Since, we are dealing with DC equivalent circuit [DC analysis] all capacitors act as open circuits to d.c signal.
Consider, the illustration below;



(All connections connecting capacitors are removed)

Now, The DC equivalent circuits is;



✓ Now equations can be designed as expressions for Q - point.

$$V_2 = \frac{V_{CC} R_2}{R_1 + R_2} \quad [\text{By applying potential divider theorem}]$$

Applying KVL to input loop.

$$V_2 = V_{BE} + V_E$$

$$V_2 = V_{BE} + I_E R_E$$

$$I_E R_E = V_2 - V_{BE}$$

Thus,

$$I_E = \frac{V_2 - V_{BE}}{R_E}$$

Now,

$$\therefore I_E = I_{CQ} = \frac{V_2 - V_{BE}}{R_E}$$

$$\text{And, } V_{CEQ} = V_{CC} - I_C (R_C + R_E)$$

